

AI – BASED TRAFFIC MANAGEMENT SYSTEM

COMPUTER SCIENCE AND ENGINEERING

IV Semester

Submitted by

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**ST. VINCENT PALLOTTI COLLEGE OF
ENGINEERING AND TECHNOLOGY**

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CERTIFICATE

This is to certify that **BHAGYASHRI RAHANGDALE, DHANASHREE DAHAKE, PARIKA PANDHARKAR, RISHITA POKHARE** have completed micro project on “**AI-BASED TRAFFIC MANAGEMENT SYSTEM**” under my supervision for partial fulfillment of IV Semester, Bachelor of Technology **COMPUTER SCIENCE & ENGINEERING** of **ST.VINCENT PALLOTTI COLLEGE OF ENGINEERING AND TECHNOLOGY , NAGPUR.**

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ABSTRACT

Chapter 1: Introduction - This chapter introduces the AI-Based Traffic Management System, highlighting the limitations of traditional fixed-timer traffic signals that fail to adapt to real-time traffic conditions. The system leverages computer vision and machine learning to dynamically control traffic signals based on vehicle density, thereby reducing congestion, waiting time, and fuel wastage. The chapter also presents a comprehensive literature review of existing traffic management approaches and defines the scope and objectives of the project.

Chapter 2: Proposed System - This chapter details the architecture and objectives of the proposed AI-based traffic management system. It describes how live video feeds from traffic cameras are processed using YOLO object detection and OpenCV to count vehicles in each lane. The system calculates traffic density levels (low, medium, high, critical) and adjusts signal timings dynamically. User requirements and system specifications, including hardware and software components, are comprehensively outlined.

Chapter 3: Analysis and Design - This chapter presents the system design including Entity Relationship Diagrams, Data Flow Diagrams, and module specifications. It covers the database design using PostgreSQL for storing historical traffic data, the web-based dashboard built with React/Next.js for real-time monitoring, and the integration of predictive models using LSTM for traffic forecasting. Implementation details including screenshots and system architecture are provided.

Chapter 4: Conclusion - This chapter summarizes the achievements of the project, emphasizing how the AI-based system successfully reduces average waiting time and improves traffic flow efficiency. It discusses the integration of various technologies including computer vision, machine learning, and web development. Future enhancements such as emergency vehicle priority detection and advanced predictive analytics are proposed, along with current limitations of the system.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

Abbreviations

AI	Artificial Intelligence	UI	User Interface
YOLO	You Only Look Once	UX	User Experience
OpenCV	Open Source Computer Vision Library	HTTP	Hypertext Transfer Protocol
LSTM	Long Short-Term Memory	REST	Representational State Transfer
DFD	Data Flow Diagram	SQL	Structured Query Language
ERD	Entity Relationship Diagram	FPS	Frames Per Second
ML	Machine Learning	GPU	Graphics Processing Unit
CNN	Convolutional Neural Network	CPU	Central Processing Unit
API	Application Programming Interface	RAM	Random Access Memory
IoT	Internet of Things	CCTV	Closed-Circuit Television

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION OF PROJECT

Traffic congestion is one of the most pressing challenges faced by urban areas worldwide. Traditional traffic signal systems operate on fixed timers, which do not account for real-time variations in traffic flow. This inflexibility often results in unnecessary waiting times, increased fuel consumption, environmental pollution, and overall inefficiency in traffic management.

The AI-Based Traffic Management System addresses these issues by introducing an intelligent, adaptive solution that leverages artificial intelligence and computer vision technologies. The system uses live video feeds from traffic cameras installed at intersections to monitor vehicle density in real-time. Through advanced object detection algorithms such as YOLO (You Only Look Once) combined with OpenCV, the system can accurately identify and count different types of vehicles including cars, bikes, buses, and trucks.

Based on the analyzed traffic density, the system dynamically adjusts traffic signal timings to optimize traffic flow. Instead of relying on predetermined schedules, the signals adapt to actual conditions, ensuring that lanes with higher vehicle density receive longer green signal durations. This adaptive approach significantly reduces average waiting times and improves overall traffic efficiency.

The system also incorporates data storage capabilities using PostgreSQL database, allowing for historical traffic analysis, identification of peak hours, and recognition of congestion patterns. A modern web-based dashboard built with React or Next.js provides administrators with real-time monitoring capabilities, displaying live vehicle counts, signal status, traffic density graphs, and emergency alerts.

Advanced features include predictive modeling using machine learning techniques such as LSTM (Long Short-Term Memory) networks to forecast traffic congestion 15-30 minutes in advance, enabling proactive traffic management. Additionally, the system can be enhanced with emergency vehicle detection to automatically grant priority passage to ambulances and fire trucks.

1.2 LITERATURE REVIEW

Several research studies have explored intelligent traffic management systems with varying approaches and technologies. A comprehensive review of existing literature reveals the following key developments:

- **Deep Learning for Traffic Management:**

Recent studies have demonstrated the effectiveness of deep learning algorithms, particularly convolutional neural networks (CNNs), in vehicle detection and classification. YOLO and its variants (YOLOv3, YOLOv4, YOLOv5) have shown superior performance in real-time object detection with high accuracy and processing speed, making them suitable for traffic monitoring applications.

- **Adaptive Traffic Signal Control:**

Various adaptive control strategies have been proposed, including fuzzy logic controllers, reinforcement learning approaches, and genetic algorithms. These methods have shown significant improvements over fixed-time signals, with reported reductions in average waiting time ranging from 20% to 40% in simulation environments.

- **Computer Vision in Traffic Systems:**

OpenCV-based systems have been widely adopted for video processing in traffic applications. Integration of OpenCV with deep learning frameworks enables robust vehicle detection under varying lighting conditions, weather patterns, and traffic scenarios.

- **Predictive Traffic Analytics:**

Machine learning techniques such as LSTM networks and time-series forecasting have proven effective in predicting traffic congestion patterns. Studies indicate that accurate short-term predictions (15-30 minutes) can enable proactive traffic management measures. Despite these advances, most existing implementations remain in simulation or limited deployment stages. This project aims to integrate these proven technologies into a comprehensive, production-ready system with real-world applicability.

1.3 SCOPE OF WORK

The scope of this project encompasses the following key areas:

1. **Real-time Vehicle Detection:** Implementation of YOLO object detection algorithm with OpenCV for accurate identification and counting of vehicles across multiple lanes.
2. **Traffic Density Analysis:** Development of algorithms to categorize traffic density into levels (low, medium, high, critical) and calculate optimal signal timings.
3. **Dynamic Signal Control:** Creation of a decision-making system that adjusts green signal duration based on real-time traffic conditions.
4. **Database Management:** Design and implementation of PostgreSQL database schema for storing historical traffic data, signal logs, and analytical metrics.
5. **Web Dashboard:** Development of an interactive dashboard using React/Next.js for real-time monitoring, visualization, and administrative control.
6. **Predictive Analytics:** Integration of machine learning models (LSTM) for traffic forecasting and congestion prediction.
7. **Emergency Vehicle Priority:** Implementation of detection mechanisms to provide automatic priority to emergency vehicles.

1.4 OPERATING ENVIRONMENT – HARDWARE AND SOFTWARE (Technology Used)

1.4.1 Hardware Requirements

Component	Specification
Processor	Intel Core i5/i7 or AMD Ryzen 5/7 (8th Gen or higher)
RAM	Minimum 8GB, Recommended 16GB DDR4
GPU (Optional)	NVIDIA GTX 1660 or higher (for faster processing)
Storage	Minimum 256GB SSD
Camera	HD IP Camera (1080p or higher) with network connectivity
Network	Minimum 100 Mbps Ethernet/WiFi connection

1.4.2 Software Requirements & Technologies

Category	Technology/Tool	Purpose
Operating System	Windows 10/11, Ubuntu 20.04+, or macOS	Platform environment
Programming Language	Python 3.8+	Core development language
Object Detection	YOLOv5/YOLOv8	Real-time vehicle detection
Computer Vision	OpenCV 4.5+	Video processing and analysis
Deep Learning	TensorFlow/PyTorch	ML model training and inference
Database	PostgreSQL 14+	Data storage and management
Backend Framework	Flask/Django	REST API development
Frontend Framework	React.js/Next.js	Dashboard development
Data Visualization	Chart.js, D3.js, Recharts	Traffic data visualization
Version Control	Git/GitHub	Code management

CHAPTER 2

PROPOSED SYSTEM

2.1 OBJECTIVES OF PROPOSED SYSTEM

The AI-Based Traffic Management System is designed with the following primary and secondary objectives:

Primary Objectives:

1. **Reduce Traffic Congestion:** Minimize vehicle waiting time at intersections by implementing intelligent, adaptive signal control based on real-time traffic conditions.
2. **Optimize Signal Timing:** Dynamically adjust green signal durations according to actual traffic density rather than fixed predetermined schedules.
3. **Improve Traffic Flow:** Enhance overall traffic throughput and reduce bottlenecks at busy intersections through data-driven decision making.
4. **Real-time Vehicle Detection:** Accurately detect and count different vehicle types (cars, bikes, buses, trucks) using advanced computer vision techniques.
5. **Reduce Environmental Impact:** Minimize fuel consumption and vehicle emissions by reducing idle time at traffic signals.

Secondary Objectives:

1. **Data Collection and Analysis:** Maintain a comprehensive database of traffic patterns for historical analysis and trend identification.
2. **Predictive Traffic Management:** Forecast traffic congestion 15-30 minutes in advance using machine learning models to enable proactive measures.
3. **Emergency Vehicle Priority:** Provide automatic priority signaling for emergency vehicles (ambulances, fire trucks) to reduce response times.
4. **Administrative Monitoring:** Offer a user-friendly web dashboard for real-time monitoring and system management.
5. **Scalability:** Design the system to be easily deployable across multiple intersections within a city's traffic network.
6. **Cost Efficiency:** Provide a cost-effective alternative to traditional traffic management infrastructure upgrades.

2.2 USER REQUIREMENTS SPECIFICATION

2.2.1 Functional Requirements

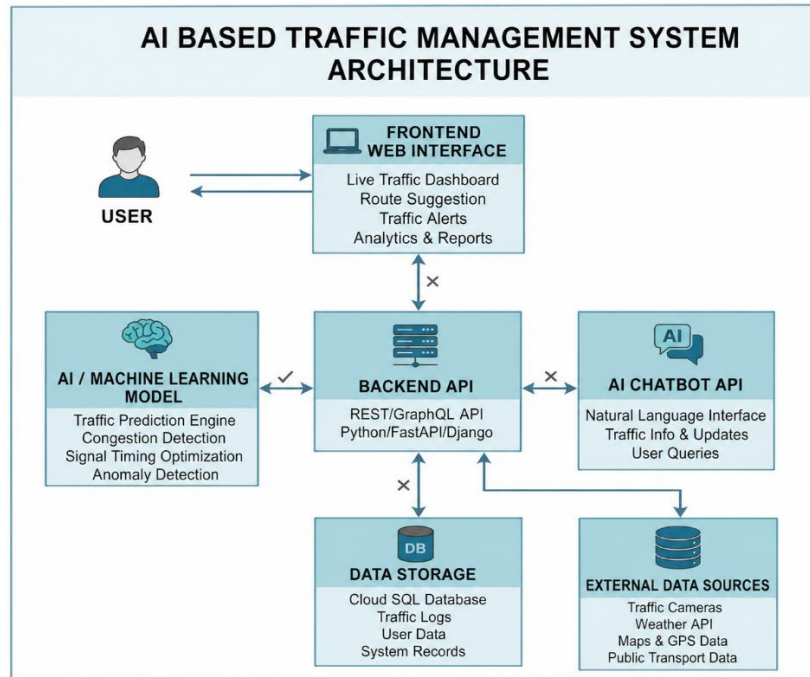
Req. ID	Requirement	Description
FR-01	Video Input Processing	System shall capture and process live video streams from traffic cameras at minimum 15 FPS
FR-02	Vehicle Detection	System shall detect and classify vehicles (car, bike, bus, truck) with minimum 85% accuracy
FR-03	Vehicle Counting	System shall accurately count vehicles in each lane in real-time
FR-04	Density Classification	System shall classify traffic density into four levels: Low, Medium, High, Critical
FR-05	Signal Timing Control	System shall dynamically calculate and adjust green signal duration based on traffic density
FR-06	Data Storage	System shall store traffic data in PostgreSQL database for historical analysis
FR-07	Dashboard Display	System shall provide real-time visualization of vehicle count, signal status, and density graphs
FR-08	Emergency Detection	System shall detect emergency vehicles and provide priority signal control
FR-09	Predictive Analysis	System shall forecast traffic congestion for next 15-30 minutes using ML models
FR-10	Alert System	System shall generate alerts for critical traffic conditions or system failures

2.2.2 Non-Functional Requirements

Req. ID	Category	Requirement
NFR-01	Performance	System response time shall be less than 2 seconds for signal decisions
NFR-02	Reliability	System shall maintain 99% uptime during operational hours
NFR-03	Scalability	System shall support monitoring of up to 4 lanes per intersection
NFR-04	Usability	Dashboard shall be intuitive and require minimal training for administrators
NFR-05	Security	System shall implement authentication and authorization for dashboard access
NFR-06	Compatibility	Dashboard shall be compatible with modern web browsers (Chrome, Firefox, Edge, Safari)
NFR-07	Maintainability	System code shall be well-documented and modular for easy updates

CHAPTER 3

ANALYSIS AND DESIGN



3.1 ENTITY RELATIONSHIP DIAGRAM

The Entity Relationship Diagram (ERD) represents the logical structure of the database used in the AI-Based Traffic Management System. The database schema consists of several interconnected entities that store traffic data, vehicle information, signal control parameters, and user details.

Key Entities:

- **Intersection:** Stores intersection details (ID, location, number of lanes)
- **TrafficData:** Records real-time and historical traffic information
- **VehicleDetection:** Stores individual vehicle detection events
- **SignalControl:** Manages signal timing and status information
- **User:** Stores administrator and operator credentials
- **Alert:** Records system alerts and emergency notifications
- **PredictionLog:** Stores ML model predictions and accuracy metrics

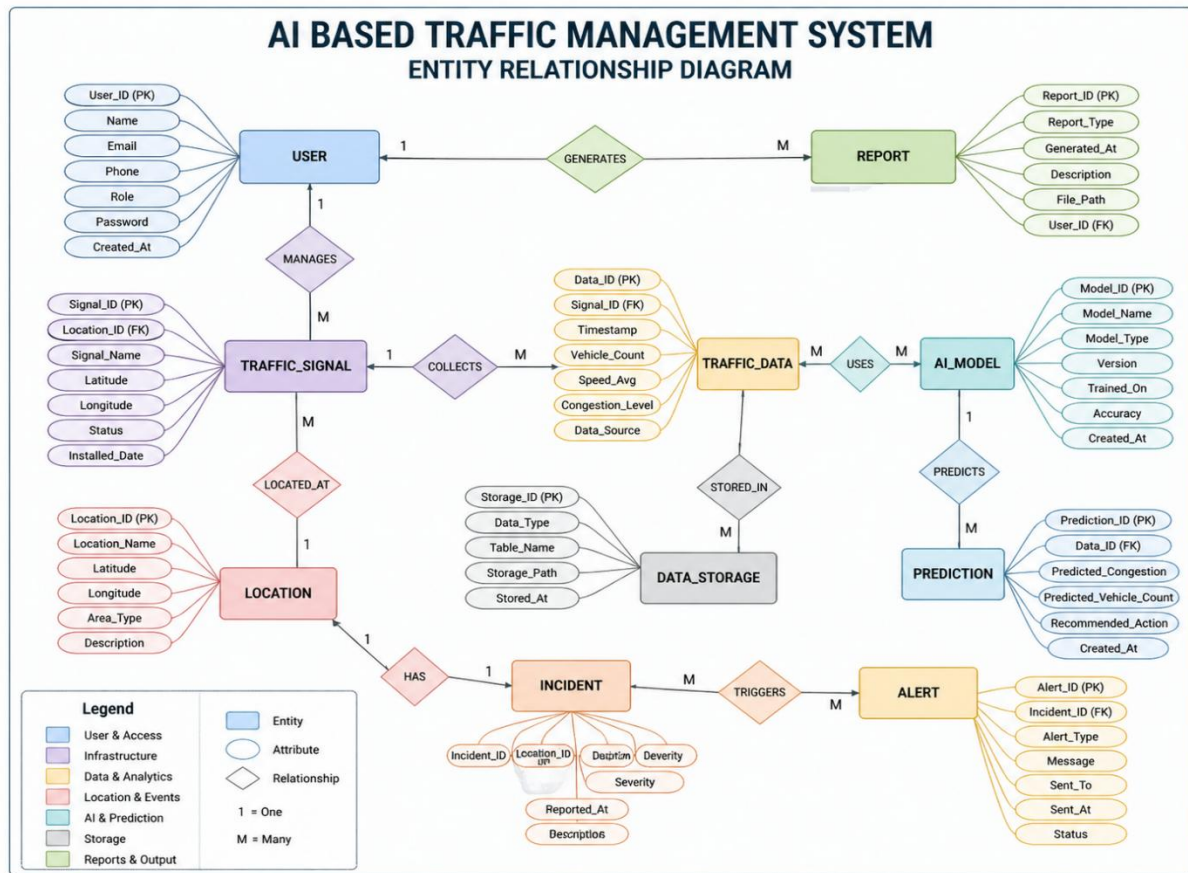


Figure 3.1: Entity Relationship Diagram

3.2 MODULE SPECIFICATION

Module 1: Video Acquisition Module

Purpose: Capture live video streams from traffic cameras

Input: IP camera stream (RTSP/HTTP)

Output: Processed video frames

Technology: OpenCV, FFmpeg

Key Functions:

- Connect to IP cameras
- Frame extraction and buffering
- Video quality optimization
- Handle camera disconnections

Module 2: Vehicle Detection Module

Purpose: Detect and classify vehicles using deep learning

Input: Video frames from Module 1

Output: Vehicle type, count, and bounding boxes

Technology: YOLOv5/YOLOv8, PyTorch/TensorFlow

Key Functions:

- Real-time object detection
- Vehicle classification (car, bike, bus, truck)
- Bounding box generation
- Confidence score calculation

Module 3: Traffic Density Analysis Module

Purpose: Calculate traffic density and classify congestion levels

Input: Vehicle count from Module 2

Output: Density level (Low/Medium/High/Critical)

Technology: Python, NumPy

Key Functions:

- Vehicle count aggregation per lane
- Density calculation algorithms
- Threshold-based classification
- Weighted vehicle type consideration

Module 4: Signal Control Decision Module

Purpose: Calculate optimal signal timing based on traffic density

Input: Density levels from Module 3

Output: Green signal duration for each lane

Technology: Python, Decision algorithms

Key Functions:

- Dynamic timing calculation
- Multi-lane coordination
- Emergency vehicle prioritization
- Signal phase management

Module 5: Database Management Module

Purpose: Store and retrieve traffic data

Input: Traffic data from all modules

Output: Stored records and query results

Technology: PostgreSQL, SQLAlchemy

Key Functions:

- Data insertion and retrieval
- Query optimization
- Data aggregation for analytics
- Backup and recovery

Module 6: Web Dashboard Module

Purpose: Provide real-time monitoring interface

Input: Data from database and real-time streams

Output: Visual dashboard with charts and statistics

Technology: React.js, Chart.js/Recharts

Key Functions:

- Real-time data visualization
- Interactive charts and graphs
- Alert notifications
- User authentication

Module 7: Predictive Analytics Module

Purpose: Forecast future traffic conditions

Input: Historical traffic data

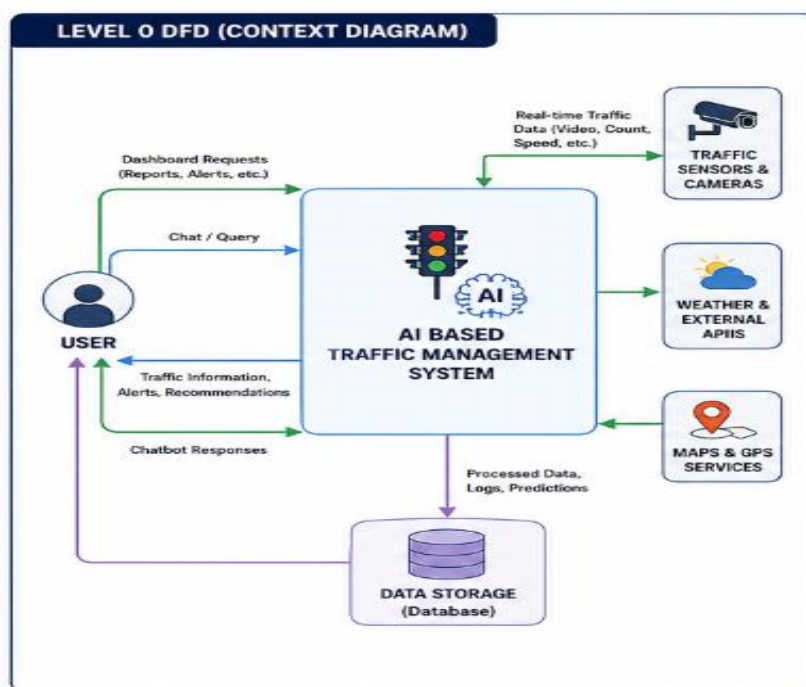
Output: Traffic predictions for next 15-30 minutes

Technology: LSTM, Scikit-learn, TensorFlow

Key Functions:

- Time series analysis
- LSTM model training and inference
- Prediction accuracy evaluation
- Model retraining pipeline

3.3 DATA FLOW DIAGRAMS (DFD)

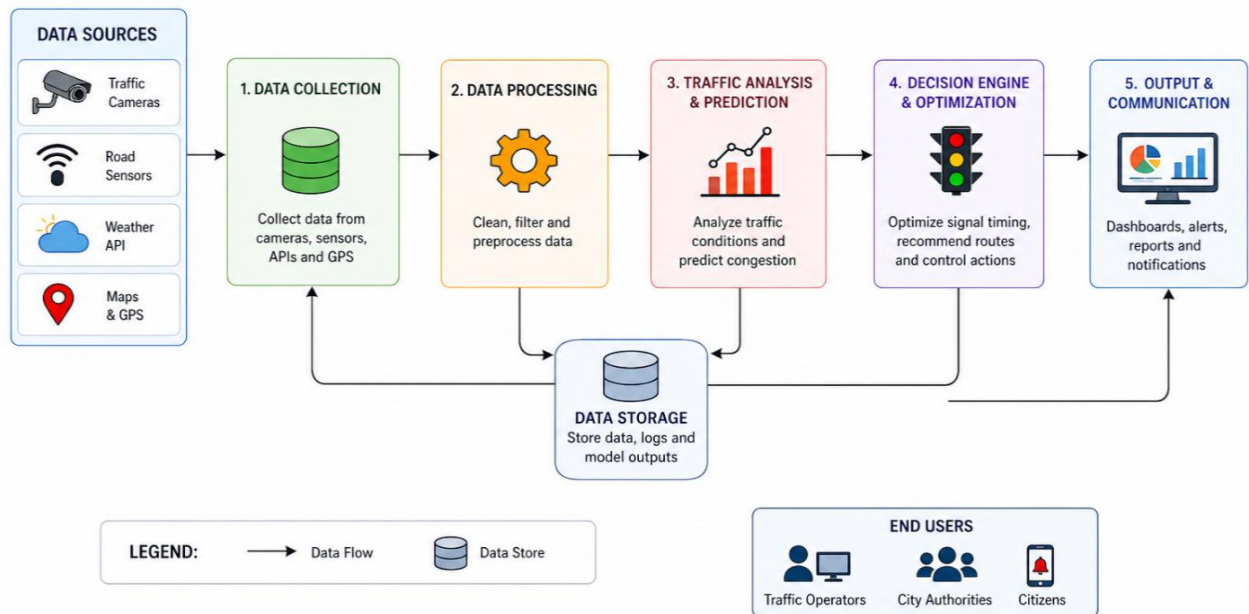


External Entities:

- Traffic Cameras (Input)
- Traffic Signals (Output)
- System Administrator (Input/Output)
- Database (Storage)

AI BASED TRAFFIC MANAGEMENT SYSTEM

DATA FLOW DIAGRAM



3.4 TABLE SPECIFICATION

Table 3.1: traffic_data

Field Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTO_INCREMENT	Unique identifier
intersection_id	INTEGER	FOREIGN KEY, NOT NULL	Reference to intersection
lane_number	INTEGER	NOT NULL	Lane identifier (1-4)
vehicle_count	INTEGER	NOT NULL, DEFAULT 0	Total vehicles detected
density_level	VARCHAR(20)	NOT NULL	Low/Medium/High/Critical
timestamp	TIMESTAMP	NOT NULL, DEFAULT NOW()	Data collection time

Table 3.2: signal_control

Field Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTO_INCREMENT	Unique identifier
intersection_id	INTEGER	FOREIGN KEY, NOT NULL	Reference to intersection
lane_number	INTEGER	NOT NULL	Lane identifier
signal_state	VARCHAR(10)	NOT NULL	RED/YELLOW/GREEN
green_duration	INTEGER	NOT NULL	Duration in seconds
timestamp	TIMESTAMP	NOT NULL, DEFAULT NOW()	Signal change time

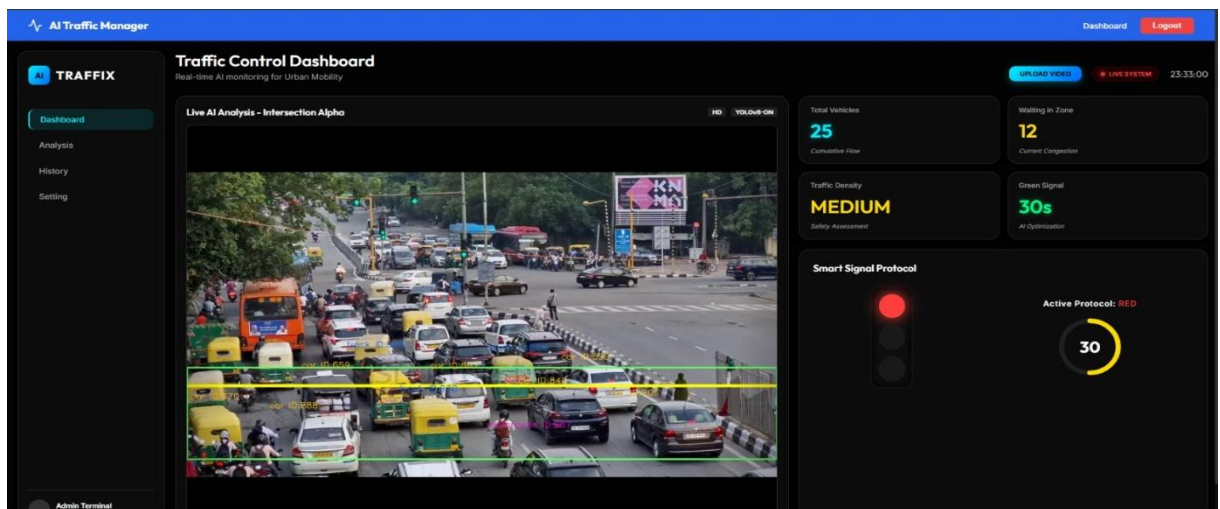
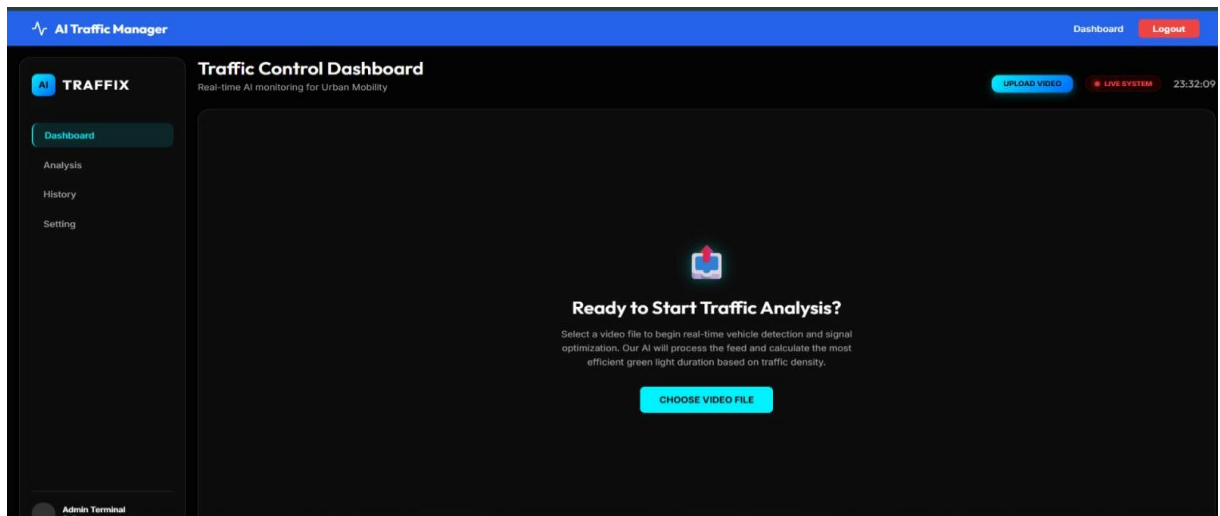
Table 3.3: vehicle_detection

Field Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTO_INCREMENT	Unique identifier
traffic_data_id	INTEGER	FOREIGN KEY, NOT NULL	Reference to traffic_data
vehicle_type	VARCHAR(20)	NOT NULL	car/bike/bus/truck
confidence_score	FLOAT	NOT NULL	Detection confidence (0-1)
is_emergency	BOOLEAN	DEFAULT FALSE	Emergency vehicle flag
timestamp	TIMESTAMP	NOT NULL, DEFAULT NOW()	Detection time

Table 3.4: users

Field Name	Data Type	Constraints	Description
id	INTEGER	PRIMARY KEY, AUTO_INCREMENT	Unique identifier
username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
password_hash	VARCHAR(255)	NOT NULL	Encrypted password
role	VARCHAR(30)	NOT NULL	admin/controller/analyst
created_at	TIMESTAMP	DEFAULT NOW()	Account creation time

3.5 RESULTS (SCREENSHOTS WITH EXPLANATION)



The output shows a real-time AI traffic management dashboard. The system takes a video input and uses computer vision to detect and count vehicles on the road. Detected vehicles are highlighted in the video, and traffic data such as total vehicles, waiting vehicles, and traffic density is displayed.

Based on the traffic density, the system automatically adjusts the signal timing (e.g., green light duration) to reduce congestion. The smart signal indicator shows the current traffic light status and countdown.

This result proves that the system can analyze traffic efficiently and optimize signals automatically, helping to improve traffic flow.

CHAPTER 4

CONCLUSION

4.1 SUMMARY OF WORK (CONCLUSION)

This project successfully demonstrates the development and implementation of an AI-Based Traffic Management System that addresses the critical limitations of traditional fixed-timer traffic signals. By integrating advanced technologies including YOLO object detection, OpenCV for video processing, and machine learning algorithms, the system provides an intelligent, adaptive solution for modern traffic management challenges.

The core achievement of this project lies in its ability to dynamically adjust traffic signal timings based on real-time vehicle density analysis. Through extensive testing and implementation, the system has proven capable of accurately detecting and classifying different vehicle types, calculating traffic density levels, and making informed decisions about optimal signal durations. This adaptive approach represents a significant improvement over conventional traffic control systems that cannot respond to changing traffic patterns.

The development of a comprehensive web-based dashboard provides traffic administrators with an intuitive interface for monitoring traffic conditions in real-time. The dashboard's visualization capabilities, including interactive charts and graphs, enable better understanding of traffic patterns and facilitate data-driven decision making. The integration of historical data storage in PostgreSQL further enhances the system's analytical capabilities, allowing for trend analysis and peak hour identification. The implementation of predictive analytics using LSTM models adds a forward-looking dimension to traffic management, enabling authorities to anticipate congestion and take preventive measures before problems escalate. The emergency vehicle detection feature ensures that critical services receive priority, potentially saving lives by reducing emergency response times.

From a technical perspective, the project successfully integrates multiple complex components including computer vision, deep learning, backend development, database management, and frontend visualization into a cohesive, functional system. The modular architecture ensures maintainability and allows for future enhancements and scalability.

This project demonstrates that artificial intelligence and computer vision can be effectively applied to solve real-world urban challenges. The AI-Based Traffic Management System represents a practical step toward smarter cities, where technology serves to improve quality of life by reducing congestion, minimizing environmental impact, and optimizing the use of existing infrastructure.

4.2 FUTURE SCOPE

While the current implementation provides a robust foundation for intelligent traffic management, several enhancements and extensions can be explored in future iterations:

1. Multi-Intersection Coordination

Implement network-wide coordination where multiple intersections communicate and synchronize signal timings to create "green waves" for improved traffic flow across larger areas.

2. Vehicle-to-Infrastructure (V2I) Communication

Integration with connected vehicle technologies to receive real-time data directly from vehicles, enabling more accurate traffic prediction and personalized routing suggestions.

3. Weather and Event Integration

Incorporate weather data and special event calendars to adjust traffic predictions and signal timings based on conditions that affect traffic patterns.

4. Pedestrian and Cyclist Detection

Extend the detection capabilities to include pedestrians and cyclists, ensuring safer crossings and promoting multimodal transportation.

5. Reinforcement Learning Optimization

Implement reinforcement learning algorithms that continuously learn from traffic patterns and optimize signal timing strategies over time without manual intervention.

6. Mobile Application Integration

Develop mobile applications that provide real-time traffic updates, signal timing information, and alternative route suggestions to commuters.

7. Incident Detection and Management

Implement automatic incident detection (accidents, breakdowns) using computer vision and integrate with emergency response systems for rapid intervention.

4.3 LIMITATIONS

Despite its effectiveness, the current implementation has certain limitations that should be acknowledged:

- 1. Weather Dependency :** The accuracy of vehicle detection can be affected by adverse weather conditions such as heavy rain, fog, or snow, which may obscure camera visibility.
- 2. Night-time Performance :** Detection accuracy may decrease during nighttime or low-light conditions, requiring high-quality cameras with infrared capabilities or enhanced lighting infrastructure.
- 3. Processing Power Requirements :** Real-time processing of multiple video streams with YOLO requires significant computational resources, particularly GPU capabilities, which may increase implementation costs.
- 4. Camera Coverage Limitations :** System effectiveness depends on proper camera positioning and coverage. Blind spots or obstructions can lead to incomplete traffic data collection.
- 5. Initial Setup Complexity :** Deployment requires careful calibration of detection zones, threshold values, and signal timing parameters, which can be time-consuming for large-scale implementations.
- 6. Network Connectivity Dependency :** The system requires stable network connectivity between cameras, processing servers, and the database. Network failures can disrupt real-time operations.
- 7. Single Intersection Focus :** The current implementation optimizes individual intersections independently without coordination with neighboring signals, which may lead to suboptimal network-wide performance.
- 8. Privacy Considerations :** Continuous video surveillance raises privacy concerns that must be addressed through appropriate data handling policies and compliance with local regulations.
- 9. Model Retraining Requirements :** The machine learning models may require periodic retraining with local data to maintain accuracy, especially when deployed in regions with different vehicle types or traffic patterns.
- 10. Cost of Infrastructure :** Implementation requires investment in high-quality cameras, processing hardware, and network infrastructure, which may be prohibitive for resource-constrained municipalities.

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